

Cumulative Distribution function

$$\text{PMF } P_X(x_k) = 1/4$$

PMF discrete random variable

CDF: can be used for any kind of random variable

Definition:

A CDF of random variable X is defined as

$$F_X(x) = P(X \leq x) \text{ for all } x \in \mathbb{R}$$

Example: I have tossed a coin twice. RV X is the number of heads. find its PMF and CDF

HH — 2

HT — 1

TH — 1

TT — 0

$$R_X = \{0, 1, 2\}$$

$$P_X(0) = 1/4 \quad P_X(1) = 2/4 \quad P_X(2) = 1/4$$

PMF

CDF: $F_X(x) = P(X \leq x)$

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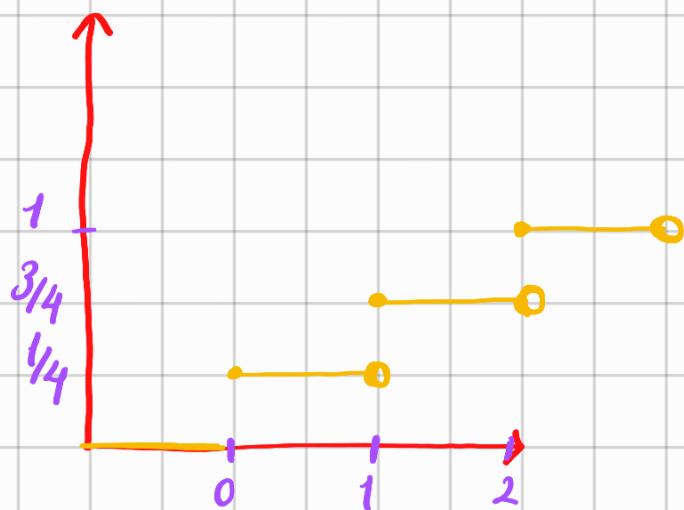
$$\text{for } \alpha < 0 \quad F_X(\alpha) = 0$$

$$\alpha \geq 2 \quad F_X(\alpha) = 1 \quad \longrightarrow \quad P_X(0) + P_X(1) + P_X(2) = \frac{1}{4} + \frac{2}{4} + \frac{1}{4} = 1$$

$$\text{For } 0 \leq \alpha < 1 \quad F_X(\alpha) = P_X(0) = \frac{1}{4}$$

$$\text{For } 1 \leq \alpha < 2 \quad F_X(\alpha) = P(X \leq \alpha) = P_X(0) + P_X(1) = \frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

$$F_X(\alpha) = \begin{cases} 0 & \alpha < 0 \\ \frac{1}{4} & 0 \leq \alpha < 1 \\ \frac{3}{4} & 1 \leq \alpha < 2 \\ 1 & \alpha \geq 2 \end{cases}$$



Properties of Cumulative Dist. Function

$$\textcircled{1} F_X(-\infty) = 0 \quad F_X(\infty) = 1$$

CDF'nin deperi
her zaman ortar

$\textcircled{2}$ CDF is a non-decreasing function

$$\alpha \leq \beta \implies F_X(\alpha) \leq F_X(\beta)$$

$\textcircled{3}$ For any $x_k \in \mathbb{R}_X$

$$F_X(x_k) - F_X(x_k - \epsilon) = P(x = x_k)$$

$\hookrightarrow$ Very small
value

$$\textcircled{4} P(a < x \leq b) = P(x \leq b) - P(x \leq a)$$

$$= F_X(b) - F_X(a)$$

Expectation

$$a_1, a_2, a_3, \dots, a_n$$

$\hookrightarrow$ average

$$\underbrace{\sum_{i=0}^n a_i}_{\text{mean}}$$

average

Consider a R.V X , how we find its average?

Let X be a R.V with $R_X = \{x_1, x_2, x_3, \dots\}$
the expected value of X , EX

$$EX = \sum_{x_k \in R_X} x_k \underbrace{P(X=x_k)}_{P_X(x_k)}$$

$$E[X] = E(x) = \mu_X$$

expected value

yukarıdaki ifade bazı kitaplarda böyle de gösteriliyor

example: X 'in beklenen degerini bulalım

x	$P(x)$
2	0,75
3	0,25

$$\begin{aligned} EX &= \sum x_k P_X(x_k) \\ &= 2 \cdot 0,75 + 3 \cdot 0,25 \\ &= 1,5 + 0,75 \\ &= 2,25 \quad \mu_X \end{aligned}$$

Functions of random variables

X is a random variable

$$Y = g(X)$$

random variable

function

R.V

yani bir rastgele deęistenin fonksiyonları bunlar girdi olarak rastgele bir deęisten alır ve bunlar yine rastgele bir deęisten olan Y üretir

$$R_Y = \{ g(x) \mid x \in R_X \}$$

range

We already know the $PMF(x)$,
biz $PMF(x)$ 'i biliyoruz o zaman $PMF(y)$ nasıl elde ederiz

$$P_Y(y) = P(Y=y) = P(g(X)=y) = \sum_{x: g(x)=y} P_X(x)$$

Ex: Let X be a discrete RV with $P_X(k) = 1/5$
for $k = -1, 0, 1, 2, 3$

Let $Y = 2|X|$ find the range and PMF of Y

$$R_X = \{-1, 0, 1, 2, 3\}$$

$$R_Y = \{2, 0, 2, 4, 6\}$$

$$R_Y = \{0, 2, 4, 6\}$$

$\rightarrow$ x 'in range'indeki
her elemanı $2|x|$ 'de
yerine koyduk

$$P_X(0) = P_X(-1) = P_X(1) = P_X(2) = P_X(3) = 1/5$$

$$P_Y(0) = P(Y=0) = (2|X|=0) = P_X(X=0) = 1/5$$

$$P_Y(2) = P_Y(2|X|=2) = P_X(1) + P_X(-1) = 1/5 + 1/5 = 2/5$$

$\begin{array}{c} \swarrow \quad \searrow \\ +1 \quad -1 \end{array}$

$$P_Y(4) = P_Y(2|X|=4) = P_X(2) = 1/5$$

$\swarrow$
2

$$P_X(6) = 1/5$$

Law of unconscious statistician (Lotus)
for discrete random variable

$$\mathcal{E}[\underbrace{g(x)}_Y] = \sum_{x_L \in R_X} g(x_L) P_X(x_L)$$

(for proof see your book)

Ex: Prev question find $E[Y]$ where $Y = 2|X|$

$$R_X = \{-1, 0, 1, 2, 3\}$$

Lotus

$$\begin{aligned} \mathcal{E}[Y] &= 2|-1| \cdot P_X(-1) + 2|0| \cdot P_X(0) + 2|1| \cdot P_X(1) \\ &\quad + 2|2| \cdot P_X(2) + 2|3| \cdot P_X(3) \end{aligned}$$

$$\begin{aligned} &\rightarrow 2 \cdot 1/5 + 0 + 2 \cdot 1/5 + 4 \cdot 1/5 + 6 \cdot 1/5 \\ &= \mu_{E(Y)} \end{aligned}$$

$$2 + 2 + 4 + 6 = 14$$

$P(X \leq x)$
Cumulative distri-

5

5

bution function

$\mu \rightarrow$ average of random variable

Variance and standard deviation

I am offered to invest \$800 2 investment accounts:

1- will give me \$1000 in one year

2- will give me either 500\$ or \$1500 (equally likely) in one year

which one should I choose?

$$P(X=1000)=1$$

$$EX=1000$$

expected value

$$\text{Var}(X)=1000-1000=0$$

Same

$$Y \begin{cases} 1/2 & 500 \\ 1/2 & 1500 \end{cases}$$

$$EX=500 \cdot 1/2 + 1500 \cdot 1/2 = 1000$$

Variance

Only expectation is not sufficient

Bu ikişinin expectationları aynı çıktı yani bunlar yeterli değil

$$\text{Var}(Y) = (500-1000)^2 + (1500-1000)^2$$

Something different then zero

zero

O yüzden bizim variance and standard deviation'ın ihtiyacımız var

and standard deviation is indicated for

The variance of a RV X with $EX = \mu_x$

$$\text{Var}(x) = E[(x - \mu_x)^2]$$

G^2

mean

Variance

Variance

bazı kitaplarda böyle gösterilir

G^2

variance

G

standart deviation

$\text{Var} = 0$

↳ yukarıda variance'im 0 olucak

↓

ama 500, 1500

değerlerini koydu-

ğumuzda variance > 0 olur

Burada çıkan değer expectation ya da average-dan farklı değer gösteriyor

Definition → standart deviation is a square root of variance

$$G = \text{std}(X) = \sqrt{\text{Var}(x)}$$

standart deviation

variansın kareköküdür

standart deviation

Covariance and Correlation

Ex, var, std. dev $\rightarrow$ distribution of a single RV

Two R.V. $\rightarrow$ Covariance, Correlation (iki tane random variables varsa bunları kullan)

$\rightarrow$ Covariance measure the association of two R.V

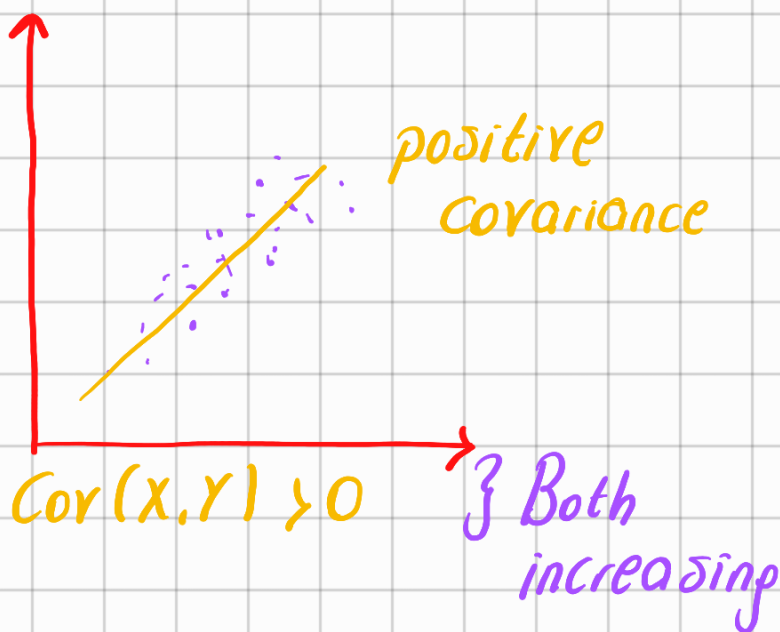
Covariance: Two R.V X and Y

$\text{Cov}(X, Y)$ is defined as =

$$= E(X - E_X), E(Y - E_Y)$$

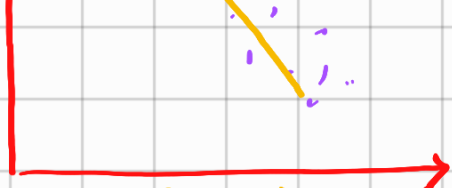
$$E(XY) - E(X) \cdot E(Y) \rightarrow \text{proof see book}$$

$\rightarrow$ Covariance formula



Covaryans 0'dan büyükse bu ikisinin de pozitif olduğu anlamına gelir





$Cov(X, Y) < 0$ → one of X, Y is increasing and the other one is decrease



Covaryans 0 ise X ve Y arasında bir ilişki yoktur
zero
Covariance

$Cor(X, Y) = 0$ → Bu X ve Y arasında herhangi bir ilişki olmadığı anlamına gelir
no relation

Correlation

Definition

Like covariance, the values of correlation is between $[-1, 1]$

$$P = \frac{Cor(X, Y)}{(\sigma_d X)(\sigma_d Y)} = \frac{Cor(X, Y)}{\sigma_x \sigma_y}$$

Bu ρ olarak okunuyor ρ harfi değil bu

↓
Sigma

eper covariance

1 or $-1 \rightarrow$ perfect correlation

1 veya -1 ise
bizi buno perfect
correlation deriz

$\text{Cor}(X, Y) < 0 \rightarrow$ small X , large Y

$\text{Cor}(X, Y) = 0 \rightarrow X$ and Y are uncorrelated

Ex: A program consists of two modules. the number of errors in the first module X and the number of errors in the second module Y have joint distribution

$P(X, Y)$

$P(0,0) = P(0,1) = P(1,0) = 0.2$ $\nearrow$ joint dist

$P(1,1) = P(1,2) = P(1,3) = 0.1$

$P(0,2) = P(0,3) = 0.05$

a.) marginal prob X and Y $P(X)$
 $P(Y)$

b.) X and Y are independent?

c.) $P(X, Y) = ?$

d.) $P(X, Y) = ?$

d-1) $\text{Var}(X) = ?$ $\text{Var}(Y) = ?$

	0	1	2	3
0	0.2	0.2	0.05	0.05
1	0.2	0.1	0.1	0.1

$P_Y(0)$ $P_Y(3)$

$$P_X(0) = \sum_y P(x, y) = 0.2 + 0.2 + 0.05 + 0.05 = 0.5$$

$$P_X(1) = 0.5$$

$$P_Y(0) = \sum_x P(x, 0) = P(0, 0) + P(1, 0) = 0.4$$

$$P_Y(1) = 0.3$$

$$P_Y(2) = 0.15$$

$$P_Y(3) = 0.15$$

b) Are X and Y independent?

$$P(X, Y) = P(X) \cdot P(Y) \longrightarrow X \text{ and } Y \text{ are independent}$$

Proof with contradiction

assume X and Y are independent

$$X=0 \text{ and } Y=1$$

$$P_{X,Y}(0,1) = P_X(0)P_Y(1)$$

$$0.2 = 0.5 \cdot 0.5$$

No they are not equal

X and Y are not independent with
probably contradiction

$$C) \text{Var}(X) = ? \quad \text{Var}(Y) = ?$$

$$\text{Var}(X) = E(X - \mu)$$

$\hookrightarrow EX$

$$E(X) = \mathcal{I}_X = \sum_{x=0}^1 x P(x) = 0 \cdot 0.5 + 1 \cdot 0.5 = 0.5$$

$$E(Y) = \mathcal{I}_Y = \sum_{y=0}^3 y P(y) = 0 \cdot 0.4 + 1 \cdot 0.3$$

$$+ 2 \cdot 0.15 + 3 \cdot 0.15$$

$\rightarrow 1.05$

$0.3 \quad \quad \quad 0.45$

$$\text{Var}(X) = E[(X - \mu_X)^2] = \sum_{x=0}^1 (0 - 0.5)^2 + (1 - 0.5)^2$$

$x=0$

$\rightarrow 0.5$

$$\text{Var}(Y) = E[(Y - \mu_Y)^2] = (0 - 1.05)^2 + (1 - 1.05)^2$$

homework

4 values

Solve it

$$d-1) \rho_{X,Y} = \text{Cor}(X,Y)$$

$$\begin{aligned}\text{Cor}(X,Y) &= E((X - \mu_X)(Y - \mu_Y)) \\ &= E(X,Y) - E(X) \cdot E(Y)\end{aligned}$$

Covariance

a, b, c $\rightarrow$ constant
properties of variances and covariances

$$\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cor}(X,Y)$$

$\swarrow \searrow$
X ve Y random variable olur

$$\begin{aligned}\text{Cor}(aX+bY, cZ+dW) &= ac \text{Cor}(X, Z) + \\ &\quad ad \text{Cor}(X, W) + \\ &\quad bc \text{Cor}(Y, Z) + \\ &\quad bd \text{Cor}(Y, W)\end{aligned}$$

$$\text{Cor}(X, Y) = \text{Cor}(Y, X)$$

$$\rho(X, Y) = \rho(Y, X)$$

in particular

$$\text{Var}(aX+b) = a^2 \text{Var}(X)$$

$$\text{Cor}(aX+b, cY+d) = ac \text{Cor}(X, Y)$$

$$\rho(aX+b, cY+d) = \rho(X, Y)$$

for independent X and Y

$$\text{Cor}(X, Y) = 0$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

4. parti kaldı



